

Differentials, Demystified

Why you *can* multiply by dx in a separable ODE, even though $\frac{dy}{dx}$ is “not a fraction”

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The paradox, stated plainly. You have been told two things that seem to contradict each other:

- (1) $\frac{dy}{dx}$ is *not* a fraction. It is a single symbol for a limit, so you are not allowed to split it into a numerator dy and a denominator dx .
- (2) Yet to solve a separable equation you write $dy = f(x) dx$, integrate both sides, and it always gives the right answer.

Both statements are correct. The resolution is that step (2) is a **shorthand**: every “illegal” move is silently invoking a genuine theorem that happens to make the notation behave like a fraction. This document expands the shorthand back into the theorems it stands for.

1 First, why the warning is literally true

By definition,

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x},$$

where $\Delta y = y(x + \Delta x) - y(x)$. The quotient $\Delta y / \Delta x$ *is* an honest fraction of two real numbers. But the derivative is the *limit* of that fraction, and a limit is a single number. There is no leftover “ dy ” or “ dx ” sitting around once the limit has been taken — they were consumed in the limiting process. So at the level of introductory calculus, dy/dx is one indivisible symbol, and the warning is right.

The interesting question is therefore not “is it a fraction?” (it is not) but “*why does treating it like one keep working?*” The answer is a single theorem.

2 The one theorem behind all of it

Theorem 2.1 (Substitution rule, i.e. the chain rule run backwards). *If $F' = f$ and u is a differentiable function of x , then*

$$\int f(u(x)) u'(x) dx = F(u(x)) + C = \int f(u) du.$$

The famous mnemonic “ $du = u'(x) dx$ ” is nothing more than a *name* for this theorem. The dx does not cancel because of algebra; it cancels because the theorem says the two integrals are equal.

Everything that looks like fraction juggling — u -substitution, separable ODEs, even the chain rule written as $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ — is this one fact wearing different clothes. Leibniz deliberately

designed the symbols so that the theorem would *look* like cancellation. That is a feature, not an accident (more on this at the end).

A second, equivalent phrasing is worth keeping in your pocket, because it makes the ODE argument airtight:

Two functions of x with the same derivative differ only by a constant.

3 Separable ODEs: the shorthand, then the expansion

A separable equation has the form

$$\frac{dy}{dx} = g(x) h(y).$$

Shorthand (the way it is usually written). Treat $\frac{dy}{dx}$ as a fraction and shuffle symbols:

$$\frac{dy}{h(y)} = g(x) dx \quad \implies \quad \int \frac{dy}{h(y)} = \int g(x) dx.$$

Two genuinely forbidden moves were used: “multiplying by dx ” and “slapping $\int$ on a loose dy .”

Expanded (what the shorthand abbreviates). No symbol is ever split. Remember that y is an unknown *function* $y(x)$, and assume $h(y) \neq 0$ on the interval of interest.

Step 1. Divide both sides by $h(y(x))$ — legal, ordinary division of numbers:

$$\frac{1}{h(y(x))} \frac{dy}{dx} = g(x).$$

Step 2. Both sides are now ordinary functions of x . If they are equal, their antiderivatives in x are equal up to a constant:

$$\int \frac{1}{h(y(x))} \frac{dy}{dx} dx = \int g(x) dx.$$

Step 3. Apply the substitution rule to the left integral, with the inner function $u = y(x)$ and $f(y) = 1/h(y)$:

$$\underbrace{\int \frac{1}{h(y(x))} \frac{dy}{dx} dx}_{\text{theorem: this equals } \int \frac{1}{h(y)} dy} = \int \frac{1}{h(y)} dy.$$

Combining the three steps gives *exactly* the shorthand’s conclusion,

$$\int \frac{1}{h(y)} dy = \int g(x) dx,$$

but nothing was ever divided into “ dy over dx .” The shorthand and the expansion produce identical work; the difference is only that the expansion names a theorem at each step instead of pretending the symbol is a fraction.

4 A worked example, side by side

Solve $\frac{dy}{dx} = xy$. Here $g(x) = x$ and $h(y) = y$.

Shorthand.

$$\begin{aligned}\frac{dy}{dx} &= x y \\ \frac{1}{y} dy &= x dx \\ \int \frac{1}{y} dy &= \int x dx \\ \ln |y| &= \frac{x^2}{2} + C \\ y &= A e^{x^2/2}\end{aligned}$$

In full detail. $y = y(x)$, and suppose $y \neq 0$:

$$\begin{aligned}\frac{1}{y(x)} y'(x) &= x \\ \frac{d}{dx} \ln |y(x)| &= \frac{d}{dx} \left(\frac{x^2}{2} \right) \\ \therefore \ln |y(x)| &= \frac{x^2}{2} + C \\ y &= A e^{x^2/2}\end{aligned}$$

In the rigorous column, the second line is the *whole trick*: by the chain rule $\frac{d}{dx} \ln |y(x)| = \frac{y'(x)}{y(x)}$, so the left side really is the derivative of $\ln |y|$. Once both sides are recognised as derivatives of known functions, “two functions with the same derivative differ by a constant” finishes the job. The shorthand’s “ $\int \frac{1}{y} dy = \ln |y|$ ” was secretly this chain-rule recognition all along.

An honest footnote the shorthand hides: dividing by y in step one quietly assumed $y \neq 0$, so it drops the constant solution $y \equiv 0$. The expanded constant $A = \pm e^C$ ranges over nonzero values; allowing $A = 0$ afterwards restores the lost solution. The careful version makes such assumptions visible.

5 The same engine drives u -substitution

Evaluate $\int 2x \cos(x^2) dx$. The shorthand sets $u = x^2$, writes “ $du = 2x dx$,” and gets

$$\int \cos u du = \sin u + C = \sin(x^2) + C.$$

Why is “ $du = 2x dx$ ” allowed? Because it is the same substitution theorem, and you can verify the result with no differentials at all: by the chain rule,

$$\frac{d}{dx} \sin(x^2) = \cos(x^2) \cdot 2x,$$

which is exactly the integrand. The “ du ” bookkeeping is just a fast way to organise this chain-rule check.

6 Exact equations: the same trick in two variables

Consider a curve defined implicitly by $F(x, y) = C$, describable locally as $y = y(x)$. Differentiating both sides looks, on its face, like more forbidden fraction algebra:

$$F_x + F_y \frac{dy}{dx} = 0 \quad \implies \quad F_x dx + F_y dy = 0,$$

where $F_x = \frac{\partial F}{\partial x}$ and $F_y = \frac{\partial F}{\partial y}$ are evaluated along the curve. The right-hand equation is exactly the form

$$M dx + N dy = 0, \quad M = F_x, \quad N = F_y,$$

that a differential-equations course calls an **exact equation**. The step from the implicit equation to this one again looks like cross-multiplying dx into the indivisible symbol $\frac{dy}{dx}$. What actually licenses it?

6.1 The honest derivation

Theorem 6.1 (Total derivative, i.e. the chain rule for two variables). *Let $F(x, y)$ have continuous partial derivatives, and let $y = y(x)$ be differentiable. Then*

$$\frac{d}{dx} F(x, y(x)) = F_x(x, y(x)) + F_y(x, y(x)) y'(x).$$

This is proved the same way the ordinary chain rule is — by bounding the error in the linear approximation of F near a point and taking a limit — with no differential ever divided or multiplied on its own¹. It is an honest statement about the single real-valued function $x \mapsto F(x, y(x))$.

Now suppose $F(x, y(x)) = C$ for all x in some interval — that is what “ $F(x, y) = C$ defines $y(x)$ ” means. Differentiating the constant function gives 0, so by the theorem,

$$F_x + F_y y'(x) = 0$$

for every such x . This is Step 1 of the separable-ODE argument again, in the same disguise: one honest equation between functions of x , obtained with the chain rule and never by splitting dy/dx .

6.2 What “ $M dx + N dy = 0$ ” really packages

Run the logic forward to see what the compact notation stands for. Define, for a function of two variables, its **total differential**

$$dF := F_x dx + F_y dy.$$

Just as $dy = \frac{dy}{dx} dx$ is a genuine identity between 1-forms (Section 8), dF is not built by dividing anything — it is a new object, a linear combination of the symbols dx and dy , defined precisely so that its restriction to a curve $y = y(x)$ recovers the chain-rule expression above:

$$dF|_{y=y(x)} = (F_x + F_y y'(x)) dx.$$

So the sentence “ $F_x dx + F_y dy = 0$ ” is shorthand for “the total derivative of F along the curve is zero,” which by the theorem above and *equal derivatives differ by a constant* (applied to the identically-zero derivative) means $F(x, y(x))$ is constant — exactly the C we started with. Nothing was cross-multiplied; the notation was engineered so that packaging the chain rule as “ $dF = 0$ ” and unpacking it back to “ $F = C$ ” are inverse operations.

¹The full ε - δ argument is carried out in the companion note *Rigorous Derivation of the Total Derivative*.

6.3 Running it the other way: recovering F from M and N

The differential-equations use of this idea starts from the other end. One is handed

$$M(x, y) dx + N(x, y) dy = 0$$

with no F in sight, and asked to solve it. The equation deserves to be called *exact* precisely when some F exists with $F_x = M$ and $F_y = N$ — i.e., when $M dx + N dy$ really is dF for some function F , so the argument above applies and the solution curves are the level sets $F(x, y) = C$.

Proposition 6.2 (Exactness test). *If $M = F_x$ and $N = F_y$ for some twice continuously differentiable F , then*

$$\frac{\partial M}{\partial y} = F_{xy} = F_{yx} = \frac{\partial N}{\partial x},$$

by equality of mixed partial derivatives. Conversely, if $M_y = N_x$ throughout a simply connected region, such an F exists.

The forward direction is Clairaut’s theorem in disguise — another “obvious” cancellation ($F_{xy} = F_{yx}$) that is really a nontrivial theorem about mixed partials, not a symbol-swap. The converse is the same existence fact behind Section 8’s 1-forms: a 1-form with zero exterior derivative is locally exact (the Poincaré lemma). The condition $M_y - N_x = 0$ is precisely the statement that $d(M dx + N dy) = 0$.

Worked example. Solve $(2xy + 3) dx + (x^2 - 1) dy = 0$. Here $M_y = 2x = N_x$, so the equation is exact. Integrate M in x , holding y fixed:

$$F(x, y) = \int (2xy + 3) dx = x^2 y + 3x + g(y).$$

Match F_y against N to pin down g :

$$F_y = x^2 + g'(y) \stackrel{!}{=} x^2 - 1 \implies g'(y) = -1 \implies g(y) = -y.$$

So $F(x, y) = x^2 y + 3x - y$, and the implicit solution is

$$x^2 y + 3x - y = C.$$

Differentiating this back with the ordinary and chain rules reproduces the original equation exactly — confirming that “cross-multiplying by dx ” and “taking d of both sides of $F = C$ ” are the same theorem viewed from opposite ends.

A caution. Do not confuse this additive total differential with the multiplicative cyclic relation of Section 7. $dF = F_x dx + F_y dy$ is a *sum*, licensed by the chain rule for a function of two variables; $(\partial x / \partial y)(\partial y / \partial z)(\partial z / \partial x) = -1$ is a *product* of three different partials of three different two-variable functions implicitly linked by one relation. No amount of total-differential bookkeeping turns one into the other. Keeping straight *which* theorem is in play is, as always, the whole game.

7 Where the fraction picture genuinely fails

The cancellation only works for a *first* derivative of *one* variable. Push it further and it bites back:

Second derivatives don’t cancel. The symbol $\frac{d^2 y}{dx^2}$ is not $(dy)^2 / (dx)^2$; you cannot treat the “2”s as exponents and simplify. It is shorthand for $\frac{d}{dx} \left(\frac{dy}{dx} \right)$.

Partial derivatives don't cancel. For three quantities linked by one relation,

$$\left(\frac{\partial x}{\partial y}\right)\left(\frac{\partial y}{\partial z}\right)\left(\frac{\partial z}{\partial x}\right) = -1,$$

not +1. Naive “cancellation” gives the wrong sign — a famous trap in thermodynamics.

This is exactly why instructors warn you off the fraction picture: it is a reliable mnemonic in a narrow setting and a source of real errors outside it. Knowing *which theorem* you are using tells you precisely when the shortcut is safe.

8 The deeper resolution: differentials really are objects

There is a more advanced viewpoint in which dx and dy are not informal shorthand at all, but well-defined mathematical objects called **differential 1-forms**. In that framework, for a function y of a single variable x , the equation

$$dy = \frac{dy}{dx} dx$$

is a genuine, provable identity — a true statement about 1-forms, not a mnemonic. Integration $\int(\cdots)dx$ becomes integration of forms, and the substitution rule becomes the natural “change of variables” for forms.

So Leibniz’s notation was not merely lucky. It anticipated, by about two centuries, a rigorous theory in which the symbols mean something and the cancellation is a theorem. Introductory courses say “not a fraction” because that theory is out of reach at that stage — but the manipulations were never wrong, only unjustified *at that level*.

In one breath. $\frac{dy}{dx}$ is a limit, not a fraction — so the warning is true. But every fraction-style manipulation in u -substitution and separable ODEs is a disguised application of the substitution rule (the chain rule backwards), together with “equal derivatives differ by a constant.” Exact equations run the same trick with the multivariable chain rule: $F_x dx + F_y dy = 0$ is shorthand for $dF = 0$, and $M_y = N_x$ is shorthand for “a potential function F exists,” which is Clairaut’s theorem and the Poincaré lemma in disguise. Leibniz’s notation is engineered so those theorems *look* like cancellation, and at the level of differential 1-forms the cancellation becomes literally true. Trust the shortcut for first derivatives in one variable, and for total differentials built additively from partial derivatives; reach for the theorem the moment you hit higher derivatives or naively cancelled partials.